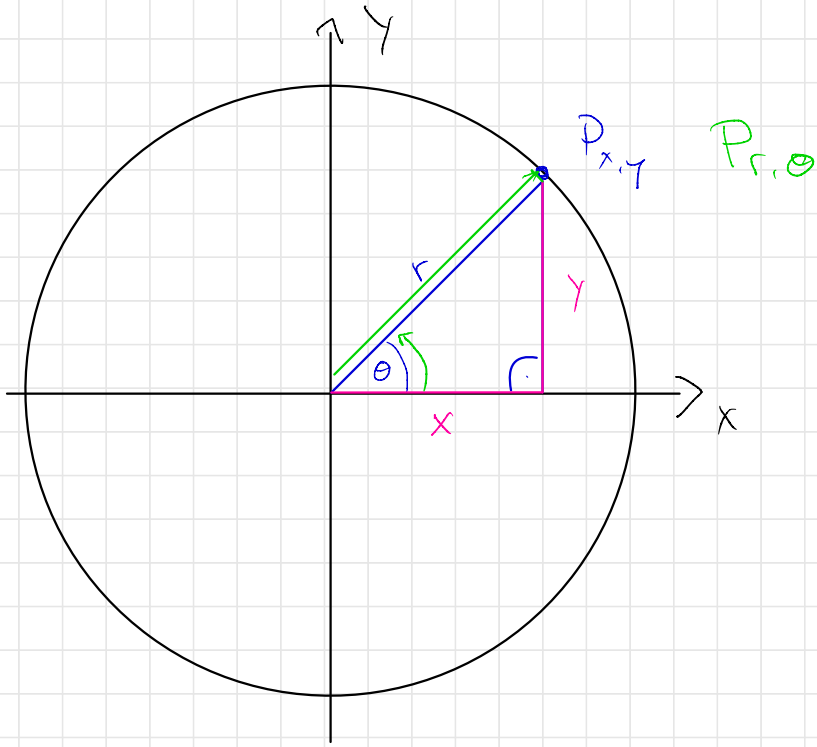
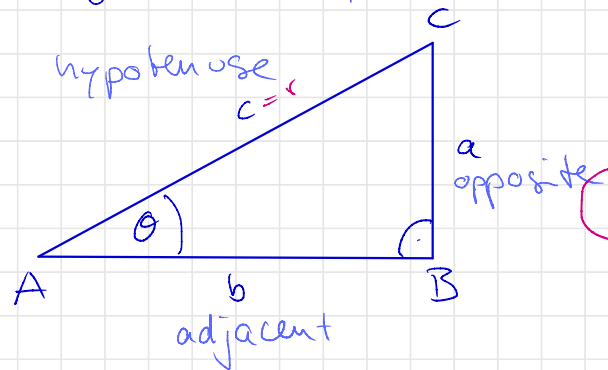


Circle Representations



Parametric Form
Trigonometry*



$$\textcircled{1} \sin(\theta) = \frac{a}{c}$$

$$\Leftrightarrow \sin(\theta) \cdot c = a$$

$$\Rightarrow y = r \cdot \sin(\theta)$$

$$\textcircled{2} \cos(\theta) = \frac{b}{c}$$

$$\Leftrightarrow \cos(\theta) \cdot c = b$$

$$\Rightarrow x = r \cdot \cos(\theta)$$

$$P_{x,y} = (r \cdot \cos(\theta), r \cdot \sin(\theta))$$

Polar Form : r, θ

From x, y to r, θ ?

① $a^2 + b^2 = c^2$ (Pythagorean theorem)

$$\Leftrightarrow \sqrt{a^2 + b^2} = c$$

$$\Rightarrow r = \sqrt{x^2 + y^2}$$

② $\tan(\theta) = \frac{a}{b}$

Apply inverse tangent on both sides

$$\Leftrightarrow \theta = \tan^{-1}\left(\frac{a}{b}\right)$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$\tan^{-1} = \text{atan}$
In praxis use atan2
instead of atan
for correct behavior
with $x < 0$ values

* Trigonometry Reminder

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b} = \frac{\frac{a}{c}}{\frac{b}{c}} = \frac{\sin \theta}{\cos \theta}$$

SOH CAH TOA

Some old horse
caught a hare
trotting over apples